Q1

What is software evolution?

Ideal Answer

Software evolution means continuously changing software from a worse state to a better state.

Q2

What is software maintenance?

Ideal Answer

Maintenance means preserving software from decline or failure.

Q3

According to Bennett and Jie Xu, what is evolution?

Ideal Answer

Evolution refers to perfective modifications triggered by changes in requirements.

Q4

What are the approaches used to study software evolution?

Ideal Answer

- Explanatory approach
- Process improvement approach

Q5

What is the explanatory approach?

Ideal Answer

It explains the causes, processes, and effects of software evolution from a scientific viewpoint.

Q6

What is process improvement approach?

Ideal Answer

It manages software evolution using better methods and tools such as maintenance, refactoring, and reengineering.

Section 2 — SPE Taxonomy

دي من أخطر أجزاء الشفوي.

Q7

What does SPE stand for?

Ideal Answer

Specified, Problem, and Evolving programs.

Q8

What is an S-type program?

Ideal Answer

An S-type program has fully and formally defined functional and non-functional requirements.

Q9

What determines the correctness of an S-type program?

Ideal Answer

Its correctness with respect to the formal specification.

Q10

Give examples of S-type programs.

Ideal Answer

- Lowest common multiple calculation
- Matrix operations

Q11

What is a P-type program?

Ideal Answer

A P-type program is based on a practical abstraction of the problem instead of a completely defined specification.

Q12

Give an example of a P-type program.

Ideal Answer

A chess-playing program.

Q13

What is an E-type program?

Ideal Answer

An E-type program is embedded in the real world and changes as the world changes.

Q14

What determines the acceptance of an E-type program?

Ideal Answer

Stakeholders' opinion and judgment.

Q15

What are the characteristics of E-type programs?

Ideal Answer

- Outcome is not definitely predictable
- Execution changes operational domain
- Evolution process is a feedback system

Section 3 — Lehman Laws

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Q16

How many Lehman laws are there?

Ideal Answer

Eight laws.

Q17

To which systems do Lehman laws apply?

Ideal Answer

E-type systems.

Q18

What is Continuing Change law?

Ideal Answer

E-type programs must be continually adapted or they become less satisfactory.

Q19

Why does complexity increase in evolving systems?

Ideal Answer

Because changes continuously add objects, modules, and subsystems.

Q20

How can increasing complexity be reduced?

Ideal Answer

By preventive maintenance.

Q21

What is preventive maintenance according to the second law?

Ideal Answer

Improving software structure without adding functionality.

Q22

What is self-regulation law?

Ideal Answer

The evolution process is self-regulating and process measures are close to normal distribution.

Q23

What is conservation of organizational stability?

Ideal Answer

The average global activity rate remains invariant during the product lifetime.

Q24

What is conservation of familiarity?

Ideal Answer

The average content of successive releases remains constant.

Q25

What is continuing growth law?

Ideal Answer

The functional content of an E-type program must continually increase to maintain user satisfaction.

Q26

What is declining quality law?

Ideal Answer

An E-type program is perceived as having declining quality if it is not maintained and adapted.

Q27

What is entropy in software?

Ideal Answer

Entropy is the increase in software complexity over time.

Q28

How can entropy be reduced?

Ideal Answer

By preventive measures improving architecture, design, and coding.

Q29

What are the types of software aging?

Ideal Answer

- Software process execution aging
- Software product aging

Q30

What are symptoms of software aging?

Ideal Answer

- Pollution
- Embedded knowledge
- Poor lexicon
- Coupling

Q31

What is code decay?

Ideal Answer

Code decay means software becomes difficult to change.

Q32

What is the eighth law?

Ideal Answer

Evolution processes are multi-agent, multi-level, multi-loop feedback systems.

Section 4 — FOSS

Q33

What does FOSS mean?

Ideal Answer

Free and Open Source Software.

Q34

Who started GNU project?

Ideal Answer

Richard Stallman.

Q35

Who started Linux?

Ideal Answer

Linus Torvalds.

Q36

What are the differences between FOSS and CSS evolution?

Ideal Answer

- Team structure
- Process
- Releases
- Global factors

Q37

Which Lehman laws do not fit large FOSS systems?

Ideal Answer

Laws 3, 4, and 5.

Section 5 — COTS & CBS

دي أكيدة جدًا.

Q38

What is a COTS component?

Ideal Answer

A deployable software component with specified interfaces and explicit context dependencies.

Q39

Why are COTS components widely used?

Ideal Answer

- Increased productivity
- Faster time to market
- Better quality

- Efficient use of human resources

Q40

What is the problem with COTS components?

Ideal Answer

Their black-box nature prevents modification and visibility into internal details.

Q41

What is wrapper?

Ideal Answer

Code used to isolate underlying components from other system components.

Q42

What is glue?

Ideal Answer

A component that combines different components together.

Q43

What is tailoring?

Ideal Answer

Enhancing the functionality of a component without modifying its source code.

Section 6 — CBS Maintenance

Q44

Why is CBS maintenance difficult?

Ideal Answer

Because of:

- Frozen functionality
- Incompatible upgrades
- Trojan horses
- Unreliable COTS
- Defective middleware

Q45

What are CBS maintenance cost drivers?

Ideal Answer

- Component reconfiguration
- Testing and debugging
- Monitoring
- Enhancing functionality
- Configuration management

Q46

What affects CBS maintainability?

Ideal Answer

- Component choice
- Architecture and integration design

Q47

What are maintainable CBS design attributes?

Ideal Answer

- Encapsulated collaborations
- Controlled interfaces
- Controlled dependencies
- Minimal coupling

- Consistent failure handling
- High visibility
- Minimal deployment effort

أهم الـ Trick Questions

Trick 1

Does preventive maintenance add functionality?

Answer

No.

Trick 2

Which systems are embedded in the real world?

Answer

E-type systems.

Trick 3

Which program type depends on formal specifications?

Answer

S-type.

Trick 4

Which program type changes with the world?

Answer

E-type.

Trick 5

Does tailoring modify source code?

Answer

No.